

A Connectivity Tax Beyond the Scalar Entropy Barrier

An exact lower certificate for the forest mode constant at $\lambda = 1$

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Abstract

Let $\mu(T)$ be the mean size of a uniformly random independent set of a finite tree T , and let $\alpha(T)$ be its independence number. Write

$$a_*(1) = \inf_T \frac{\mu(T)}{\alpha(T)},$$

where the infimum is over finite nonempty trees. Starting from the entropy-partition decomposition, we prove the strictly stronger universal bound

$$a_*(1) \geq \frac{\log(34/5)}{6 \log 2} + \frac{\log(1458/1445)}{48 \log 2} = 0.461191649955056508 \dots > 0.461.$$

The improvement comes from a connectivity tax that is absent from the scalar degree relaxation. Near scalar equality, almost all excess A -degree must be carried by vertices with $d = 4$. If the induced B -forest is dense, its edge term already pays a positive cost; if it is sparse, linearly many $d = 4$ vertices are isolated in the B -forest and can be paired through shared A -neighbours. Under the comparison hard-core measure each such pair is occupied with probability exactly $1/289$, forcing a linear overlap entropy contribution. The only scalar comparisons needed by the proof are reducible to exact integer arithmetic and are checked by the accompanying script `verify_connectivity_tax_0461.py`.

1 Statement

For a finite graph G , let

$$Z_G = |\mathcal{I}(G)|, \quad \mu(G) = \frac{1}{Z_G} \sum_{I \in \mathcal{I}(G)} |I|, \quad \alpha(G) = \max_{I \in \mathcal{I}(G)} |I|.$$

Set

$$c := \frac{\log(34/5)}{6 \log 2}, \quad \gamma := \frac{1}{2} - c = \frac{\log(20/17)}{6 \log 2},$$

and define the new separation constant

$$\rho := \frac{\log(1458/1445)}{12 \log 2} = 0.00107676891157466036 \dots$$

Theorem 1 (Connectivity-tax certificate). *Every finite nonempty tree T satisfies*

$$\frac{\mu(T)}{\alpha(T)} \geq c + \frac{\rho}{4}.$$

Consequently

$$a_*(1) \geq c + \frac{\rho}{4} = 0.461191649955056508 \dots > 0.461.$$

The proof is completely uniform in the order and maximum degree of the tree.

2 Entropy–partition decomposition

Fix a maximum independent set A of T and write

$$\alpha = |A|, \quad B = V(T) \setminus A, \quad K = T[B], \quad e = |E(K)|.$$

For $v \in B$ put

$$d_v = |N(v) \cap A|.$$

Since A is maximum, it is maximal, hence $d_v \geq 1$ for every $v \in B$. Because T is a tree,

$$\sum_{v \in B} (d_v - 1) = \alpha - 1 - e. \quad (1)$$

For $d \geq 1$ define

$$\ell_d := \left\lceil \log \frac{5/4}{1 + 2^{-d}} \right\rceil_+, \quad \Delta_d := \gamma(d - 1) - \frac{\ell_d}{2 \log 2}.$$

We shall use the following sharpened form of the scalar estimate.

Lemma 2 (Scalar separation away from $d = 4$). *One has $\Delta_1 = \Delta_4 = 0$, and for every integer $d \geq 2$, $d \neq 4$,*

$$\Delta_d \geq \rho(d - 1).$$

Proof. We have

$$\ell_3 = \log(10/9), \quad \ell_4 = \log(20/17),$$

so $\Delta_4 = 0$ and

$$\frac{\Delta_3}{2} = \gamma - \frac{\ell_3}{4 \log 2} = \frac{\log(1458/1445)}{12 \log 2} = \rho.$$

For $d \geq 2$, $d \neq 4$, the desired inequality is equivalent to

$$2\ell_d \leq (d - 1)\ell_3. \quad (2)$$

For $d = 2$ this is immediate and for $d = 3$ it is equality. For $d = 5$ it follows from

$$\left(\frac{40}{33}\right)^2 < \left(\frac{10}{9}\right)^4,$$

and for every $d \geq 6$ from

$$2\ell_d < 2 \log(5/4) < 5 \log(10/9) \leq (d - 1)\ell_3.$$

The two strict rational comparisons are checked after clearing denominators by the accompanying integer audit. $\square$

We next record exactly the part of the entropy argument that will be used. Let $J = I \cap B$ for a uniformly random independent set I of T . Put

$$w_v = 2^{-d_v}, \quad Z_K(w) = \sum_{J \in \mathcal{I}(K)} \prod_{v \in J} w_v.$$

Let Q be the hard-core measure on K with activities $(w_v)_{v \in B}$. For $J \in \mathcal{I}(K)$ define

$$O(J) := \sum_{v \in J} d_v - |N_A(J)|.$$

Then

$$O(J) = \sum_{a \in A} (k_a(J) - 1)_+, \quad k_a(J) = |J \cap N(a)|. \quad (3)$$

Counting first over J gives the exact identity

$$Z_T = 2^\alpha Z_K(w) \mathbb{E}_Q 2^{O(J)}, \quad R_A := \log \mathbb{E}_Q 2^{O(J)} \geq (\log 2) \mathbb{E}_Q O(J). \quad (4)$$

The last inequality is Jensen.

For completeness, the entropy identity with the comparison measure of activity $1/4$ is

$$2 \log 2 \mu(T) = \log Z_T - \log Z_K(1/4) + D_{\text{KL}}(P \| Q_{1/4}),$$

where P is the law of J and $Q_{1/4}(J) = 4^{-|J|}/Z_K(1/4)$. Changing the coordinates of Z_K one at a time yields

$$\log Z_K(w) - \log Z_K(1/4) \geq - \sum_{v \in B} \ell_{d_v}.$$

Indeed, with all other activities fixed the partition function is $U + xV$ with $0 < V \leq U$, so lowering a coordinate from $1/4$ to $x \leq 1/4$ costs at most $\log[(1 + 1/4)/(1 + x)]$, while increasing a coordinate costs nothing. Combining these facts and discarding the nonnegative relative entropy gives

$$\mu(T) - c\alpha \geq \gamma(1 + e) + S + \frac{R_A}{2 \log 2}, \quad S := \sum_{v \in B} \Delta_{d_v}. \quad (5)$$

Here we used (1).

3 The connectivity tax

Let

$$B_4 = \{v \in B : d_v = 4\}.$$

By Lemma 2 and (1),

$$3|B_4| \geq \alpha - 1 - e - \frac{S}{\rho}. \quad (6)$$

Let $B_4^0 \subseteq B_4$ be the vertices isolated in $K = T[B]$. At most $2e$ vertices of K are incident with an edge, hence

$$|B_4^0| \geq |B_4| - 2e. \quad (7)$$

For every $a \in A$, let

$$q_a = |N(a) \cap B_4^0|.$$

Pair arbitrarily $2\lfloor q_a/2 \rfloor$ of these neighbours. Let M be the total number of pairs produced over all $a \in A$. Since

$$\sum_{a \in A} q_a = 4|B_4^0|,$$

we have

$$\begin{aligned} M &= \sum_{a \in A} \left\lfloor \frac{q_a}{2} \right\rfloor \geq \frac{4|B_4^0| - \alpha}{2} \\ &\geq \frac{\alpha}{6} - \frac{2}{3} - \frac{14}{3}e - \frac{2}{3\rho}S. \end{aligned} \quad (8)$$

Of course $M \geq 0$ as well.

Lemma 3 (Overlap tax). *With M as above,*

$$\frac{R_A}{2 \log 2} \geq \frac{M}{578}.$$

Proof. Fix $a \in A$. If $m_a(J)$ of the selected pairs at a have both endpoints in J , then

$$m_a(J) \leq \left\lfloor \frac{k_a(J)}{2} \right\rfloor \leq (k_a(J) - 1)_+.$$

Summing over a and using (3),

$$O(J) \geq \sum_{\text{selected pairs } \{u,v\}} \mathbf{1}_{\{u,v \subseteq J\}}.$$

Every vertex in B_4^0 is an isolated vertex of K with activity $2^{-4} = 1/16$. Under Q it is therefore occupied independently with probability $1/17$. Consequently every selected pair is simultaneously occupied with probability exactly $1/289$, and linearity of expectation gives

$$\mathbb{E}_Q O(J) \geq \frac{M}{289}.$$

Now use (4). □

4 Cancellation and proof of the theorem

Set

$$\theta := 867\rho.$$

The exact scalar comparisons in the accompanying audit imply

$$0 < \theta < 1, \quad \gamma > 7\rho. \tag{9}$$

Since $M \geq 0$ and M also satisfies (8), for every $0 \leq \theta \leq 1$ we may write

$$M \geq \theta \left(\frac{\alpha}{6} - \frac{2}{3} - \frac{14}{3}e - \frac{2}{3\rho}S \right).$$

Substituting Lemma 3 into (5) and using this inequality yields

$$\begin{aligned} \mu(T) - c\alpha &\geq \gamma(1 + e) + S + \frac{\theta}{578} \left(\frac{\alpha}{6} - \frac{2}{3} - \frac{14}{3}e - \frac{2}{3\rho}S \right) \\ &= \frac{\rho}{4}\alpha + (\gamma - \rho) + (\gamma - 7\rho)e. \end{aligned}$$

The S -coefficient cancels exactly because

$$\frac{867\rho}{578} \frac{2}{3\rho} = 1.$$

By (9), the two remaining correction terms are positive. Hence

$$\mu(T) - c\alpha \geq \frac{\rho}{4}\alpha,$$

which proves Theorem 1.

5 Exact scalar audit

No floating-point comparison is needed for the theorem. The verifier `verify_connectivity_tax_0461.py` checks the following after clearing denominators:

1. the $d = 5$ comparison

$$(40/33)^2 < (10/9)^4;$$

2. the uniform $d \geq 6$ comparison

$$(5/4)^2 < (10/9)^5;$$

3. $\theta < 1$, equivalently

$$(1458/1445)^{289} < 16;$$

4. $\gamma > 7\rho$, equivalently

$$(20/17)^2 > (1458/1445)^7;$$

5. the advertised decimal threshold, equivalently

$$(34^8 \cdot 1458)^{1000} > 2^{22128} (5^8 \cdot 1445)^{1000}.$$

The last comparison is exactly

$$c + \frac{\rho}{4} > \frac{461}{1000}.$$

The verifier also checks the three rational coefficient identities responsible for the cancellation in the final display.

6 Current interval and relation to the Bellman attack

Combining Theorem 1 with the supplied exact layered-tree witness with branching factors

$$(2, 1, 3, 9, 6, 9, 9, 9, 14, 10)$$

gives the present certified interval

$$0.461191649955056508 \dots \leq a_*(1) < 0.473301418185171818.$$

The upper endpoint is a genuine integer-tree witness. It should not be confused with the separate fractional Bellman obstruction: the latter concerns a weighted relaxation and is neither a tree witness nor an upper bound on $a_*(1)$.

The significance of the present argument for the Bellman programme is structural. The old scalar entropy bound becomes sharp only at degree $d = 4$. The proof above shows that a connected tree cannot realize that scalar equality for free: either the induced B -forest pays through e , or isolated $d = 4$ vertices create overlap entropy through shared A -neighbours. Thus the first nonzero “connectivity correction” can already be extracted without solving the full Bellman dual.